

# Chapter 12

## Mean Median Mode

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

**1)** In Marian's bowling tryout, she has a 148 average after her first four games. To make her high school bowling team, she needs to average at least 150 pins in her tryout. With only one game left, **what is the lowest score** Marian can score on her **fifth** and final game to make the team?

- A) 188                      B) 154                      C) 158                      D) 166                      E) 162

**2)** The average of 9 numbers is 11. When a 10<sup>th</sup> number is added, the new average is 12. **What is the 10<sup>th</sup> number added?**

- A) 10                      B) 21                      C) 25                      D) 11                      E) 14

**3)** The average of 5 numbers is 9. If the average of two of those numbers is  $-4$ , **what is the sum of the other three numbers?**

- A) 72                      B) 53                      C) 42                      D) 21                      E) 32

**4)** The average of 6 numbers is 18. If 5 of the six numbers are; 14, 26, 21, 17, and 14, **what is the median of the six numbers?**

- A) 13                      B) 18.5                      C) 15                      D) 16.5                      E) 16

**5)** If the average of six numbers is greater than nine and less than 12, which of the following could be the **sum of the six numbers?**

- A) 54                      B) 38                      C) 60                      D) 72                      E) 76

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

**6)** If the average of 5 numbers is 13 and the average of two of those numbers is 10, **what is the average of the other three numbers?**

- A) 15                      B) 12                      C) 10                      D) 16                      E) 18

**7)** **What is the average** of  $x + 6$ ,  $3x$ ,  $4x + 2$ ,  $2x - 3$ , and 10?

- A)  $3x$                       B)  $3x + 3$                       C)  $2x$                       D)  $2x + 3$                       E)  $x + 3$

**8)** If the average of  $a$ ,  $b$ , and  $c$  is 50, and the average of  $a$ ,  $b$ ,  $c$ , and  $d$  is 70, **what is the value of  $d$ ?**

- A) 130                      B) 60                      C) 70                      D) 100                      E) 120

**9)** After four tests, **Sarah's** average math test grade is currently an 86. If there is only the final exam left and it is weighted the same as two tests, what score, if possible, does **Sarah** need on her final exam to bring her average up to **exactly a 90?**

- A) 98                      B) 95                      C) 97                      D) 100                      E) It is not possible

**10)** If  $b$  is an integer  $> 0$  and the average of  $b$ ,  $b + 3$ ,  $b^2 - 9$ ,  $-3b$ ,  $2b - 1$ , and 5 is 3. **What is the mode of the numbers?**

- A) 7                      B) 3                      C) 4                      D) 8                      E) 2

**11)** If the average of two numbers,  $g$  and  $h$ , is 0, which of the following **must be true?**

- A)  $g = 0$ ,  $h = 0$   
 B)  $gh = 0$   
 C)  $g - h = 0$   
 D)  $g + h = 0$   
 E)  $g > 0$ ,  $h < 0$

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
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|---|---|---|---|---|---|---|---|---|----|

**12)** What is the mean of the following set of numbers? **{4, 8, 12, 0, 1}**

- A) 5                      B) 6.2                      C) 5.8                      D) 4.4                      E) 7

**13)** If the mean median and mode of a set of five positive integers is 5, which of the following **must be true**?

- A) Every number in the set is a 5  
 B) The mean of the numbers in the set other than 5 is equal to 5  
 C) The median of the numbers in the set other than 5 is equal to 5  
 D) There are three or more 5's in the set of integers  
 E) The largest integer in the set must not exceed 11

**14)** If the arithmetic mean of a set of five integers is 15, and four of the integers in the set are 18, 19, 20, and 21. **What is the fifth integer in the set?**

- A) -3                      B) 11                      C) -11                      D) 15                      E) 0

**15)** If the arithmetic mean of a set of seven integers is 8, and four of the integers in the set are 8, 9, 10, and 11. **What is the mean of the other three numbers in the set?**

- A) -5                      B) 6                      C) 4                      D) 12                      E) -2

**16)** If the average of a set of numbers,  $a + 9$ ,  $3a - 3$ , 8,  $2a + 8$ , and  $4a - 2$  is  $3a - 1$ , what is the **sum** of the **median and the mode of the set**?

- A) 17                      B) 18                      C) 22                      D) 27                      E) 32

**17)** What is the average of  $x^2$ ,  $3x^2$ ,  $6x^2$ , and  $10x^2$ ?

- A)  $5x^2$                       B)  $2x^2$                       C)  $20x^2$                       D)  $45x^4$                       E)  $45x^2$



**Use the following chart for questions 18 and 19**

The chart below shows the scores for 20 students that took a particular test. The scores are based on the number of questions students answered correctly out of ten questions.

| Number of Students | Correct Answers Out of 10 Questions |
|--------------------|-------------------------------------|
| 1                  | 5                                   |
| 4                  | 6                                   |
| 2                  | 7                                   |
| 2                  | 8                                   |
| 7                  | 9                                   |
| 4                  | 10                                  |

**18)** What is the **mode** of the scores of the 20 students?

- A) 5                      B) 6                      C) 7                      D) 8                      E) 9

**19)** What is the **median** of the scores of the 20 students?

- A) 5                      B) 6                      C) 7                      D) 8                      E) 9

**20)** All of the following would not affect the median of a set of five positive integers **EXCEPT**?

- A) Adding three integers that have the same median as the original set of five integers  
 B) Adding two integers that have the same median as the original set of five integers  
 C) Duplicating the highest and lowest integer and adding them to the set  
 D) Doubling every integer greater than the median and halving every integer less than the median  
 E) Doubling every integer except the highest and lowest

|   |   |   |   |   |   |   |   |   |    |
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**21)** If the arithmetic mean of  $x$ ,  $x + 2$ ,  $2x - 4$ ,  $3x - 11$ , and  $\frac{1}{2}(x) + 3$  is 7, what is the **median** of the numbers?

- A) 6                      B) 7                      C) 8                      D) 5                      E) 12

**22)** If the average of  $a$ , 20, 40, and 80 is 45, and the average of  $b$ , 20, 40, and 80 is 60. What is the **average of  $a$  and  $b$** ?

- A) 70                      B) 52.5                      C) 55                      D) 72.5                      E) 60

**23)** If the average of 50, 70, and  $q$  is 80 and the average of 30, 60,  $q$ , and  $r$  is 85. What is **the sum of  $q$  and  $r$** ?

- A) 250                      B) 130                      C) 60                      D) 90                      E) 180

## Solutions Chapter 12

### Mean, Median, Mode

- |      |       |       |
|------|-------|-------|
| 1) C | 9) A  | 17) A |
| 2) B | 10) A | 18) E |
| 3) B | 11) D | 19) E |
| 4) D | 12) A | 20) E |
| 5) C | 13) B | 21) B |
| 6) A | 14) A | 22) A |
| 7) D | 15) B | 23) A |
| 8) A | 16) E |       |

1) When given an average, multiply the average by the number of terms. Working backward from the average formula makes these types of problems easy.

Average =  $\frac{\text{Sum}}{\text{Number of terms or subjects}}$ ; manipulate this equation and solve for the

Sum: Sum = Average x Number of subjects.

For this question, multiply the number of games played by the average to get the total number of pins (sum). Since, after four games, Nicoles's average score was 148, the sum of her scores = 4 times 148 = 592. To achieve a 150 average with five games played, multiply 150 times 5 = 750. To achieve a 150 average, Nicole needs 750 total pins. Nicole currently has 592 pins but needs a total of 750 pins. Subtract 592 from 750 = 158. If Nicole bowls a 158 in her final game, she will have a total of 750 pins in 5 games to achieve a 150 average and make her high school bowling team.

### 158 Answer C Difficulty Level 7.5

2) When given an average, work backward by multiplying the average by the number of subjects. Since nine numbers average 11; multiply 9 by 11. Those nine numbers add to 99. Since ten numbers average 12, multiply 10 by 12 = 120. So, the

ten numbers sum to 120. The 10<sup>th</sup> number added to the original set of numbers must be the difference - subtract

$$120 - 99 = 21.$$

### 21 [Answer B](#) Difficulty Level 7

3) When given the average and the number of units or trials, multiply to find the sum of the numbers. The average of 5 numbers is nine, so they must add to 5 times  $9 = 45$ . Since the average of two of the numbers is  $-4$ , multiply 2 times  $-4 = -8$ . Since all five numbers must add to 45 and two of the five numbers add to  $-8$ , the other three must add to  $45 - (-8) = 53$ . To make this clearer, since it may not be, look at the following illustration.

$$\begin{array}{c} -8 \\ \underbrace{\hspace{1cm}} \\ \{ *, *, ?, ?, ? \} \\ \underbrace{\hspace{1.5cm}} \\ 45 \end{array}$$

The five symbols preceding (2 asterisks and 3 question marks) represent the five numbers in the set. The average of two of the numbers is  $-4$ , so the two numbers (represented by two \*'s) add to  $-8$ . Since all five numbers add to 45, the 3 ???'s must add to 53 because  $53 - 8 = 45$ .

### 53 [Answer B](#) Difficulty Level 8.5

4) The average of 6 numbers is 18: Multiply  $6 \times 18 = 108$ . The six numbers add to 108. Since five of the six numbers in the set are given, sum the five numbers and subtract from 108 to find the 6<sup>th</sup> and unknown number. The numbers 14, 26, 21, 17, and 14 add to 92. Since  $108 - 92 = 16$ , then 16 is the value of the unknown number. Finally, line up the set of numbers, including 16 from least to greatest, to find the median.

~~{14, 14, 16, 17, 21, 26}~~

Since both 16 and 17 are in the middle, the median is their average/mean, 16.5.

$$\left(\frac{16+17}{2}\right) = 16.5$$

### 16.5 [Answer D](#) Difficulty Level 8

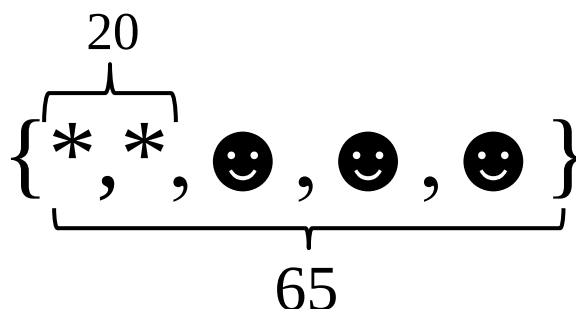
5) Average =  $\frac{\text{Sum}}{\text{Number of terms/subjects}}$ , or the

Sum = Number of Subjects x Average. If the average of six numbers equals 9 then their product/sum;  $6 \times 9 = 54$ , and if the average of six numbers equals 12 then their product/sum;  $6 \times 12 = 72$ . Therefore, the sum of the numbers must be greater than 54 and less than 72. Since 60 is the only number greater than 54 and less than 72, it is the correct answer.

### 60 [Answer C](#) Difficulty Level 7

6) If the average of five numbers is 13, multiply ( $5 \cdot 13 = 65$ ). The five numbers must add to 65. If the average of two of the numbers is ten, multiply ( $2 \cdot 10 = 20$ ). Since two of the numbers add to 20, and all five of the numbers add to 65, the sum of the other three numbers must add to 45. ( $65 - 20 = 45$ ). The average of three numbers that add to 45 is  $45/3$  or 15. The visual representation that follows may help this concept if difficult to understand. The five numbers in the set are represented by symbols, where two of the numbers are represented by asterisks, and smiley faces represent the other three numbers.

The sum of two of the numbers is 20



All five numbers add (sum) to 65

The sum of the other three numbers  $\text{☺} + \text{☺} + \text{☺} = 45$  so their average is  $45/3 = 15$  [Answer A](#) Difficulty Level 8.5

7) When taking the average of expressions containing variables (letters), do the same thing as taking the average or mean of a set of numbers. Add up all the numbers or expressions and divide them by how many there are. Notice that each of the five expressions(integers) are separated by a comma;

$$x + 6, 3x, 4x + 2, 2x - 3, \text{ and } 10$$

Add the expressions, combine like terms and divide by 5, then simplify by dividing both  $10x$  and  $15$  by 5. Many students want to cancel the  $10x$  or the  $15$  incorrectly.

$$\frac{x+6+3x+4x+2+2x-3+10}{5} = \frac{10x+15}{5} = \frac{10x}{5} + \frac{15}{5} = 2x + 3$$

$2x + 3$  [Answer D](#) Difficulty Level 8

8) If the average of  $a$ ,  $b$ , and  $c$  is 50, then the average of three numbers is 50; Multiply ( $3 \cdot 50 = 150$ ), making the sum of  $a$ ,  $b$ , and  $c$  150. For the second part of the question, repeat; four numbers  $a$ ,  $b$ ,  $c$ , and  $d$  average 70; Multiply ( $4 \cdot 70 = 280$ ) so, the sum of  $a$ ,  $b$ ,  $c$ , and  $d$  is 280.

Since  $a$ ,  $b$ , and  $c$  sum to 150 and  $a$ ,  $b$ ,  $c$ , and  $d$  sum to 280, subtract;

$(280 - 150 = 130)$ , which must be the value of  $d$ .

### 130 Answer A Difficulty Level 7.5

9) After four tests, Sarah has an average of 86. Multiply  $(4 \cdot 86 = 346)$ ; her test scores sum to 346. Since the final is worth 2 test scores, there are a total of 6 test scores. (The four Sarah already completed and the final test, which is equivalent to two test scores;  $4 + 2 = 6$ ). So, Sarah needs an average of 90 from 6 test scores. Multiply  $(6 \cdot 90 = 540)$ , which means her test scores must sum to 540. Since Sarah has 346 “points” and needs 540 “points”  $(540 - 344 = 196)$ , she needs 196 “points” on the final. Since the final is worth 2 test scores, Sarah needs 98  $(196/2 = 98)$  to finish with an overall test average of 90.

### 98 Answer A Difficulty Level 7.5

10) When solving problems on the ACT, always start by using what is given. This may sound obvious, but it works for a question like this one. There are two givens for this question; 1)  $b$  is an integer greater than zero, and 2) the average of the six expressions is 3. Use the second given and take the average of the six expressions by adding them, combining like terms, and dividing the sum by the number of expressions (6). Notice that each expression separated by a comma is an actual integer. There are six expressions,

$b, b + 3, b^2 - 9, -3b, 2b - 1,$  and 5

and their average is 3. Therefore;

$$\frac{(b + b + 3 + b^2 - 9 - 3b + 2b - 1 + 5)}{6} = 3; \frac{b^2 + b - 2}{6} = 3$$
, now multiply both sides of the equation by 6,  $b^2 + b - 2 = 18$ , then subtract 18 from both sides of the equation and set the equation equal to zero,  $b^2 + b - 20 = 0$ . Finally, factor the left side of the equation to solve. When factoring a quadratic trinomial in the form  $Ax^2 + Bx + C$  with  $A = 1$ , find two integers that add to  $B$  and multiply to  $C$ .

$B = 1$  and  $C = -20$ . Since  $5 + -4 = 1$  and  $5 \cdot -4 = -20$

factor  $b^2 + b - 20 = 0$ ;  $(b + 5)(b - 4) = 0$  and solve for  $b$ ;

$b + 5 = 0$  and  $b - 4 = 0$ . Therefore,  $b = -5$  and  $b = 4$ .

Standard form,  $Ax^2 + Bx + C$ , uses  $x$  as the variable. In this question, use  $b$  as the variable, so the equation takes the form  $Ab^2 + Bb + C$

Since the first given states, the solution must be an integer greater than 0; 4 must be the only solution. Since  $b = 4$ , find the value of each expression by substituting 4 for  $b$ ;  $b$ ,  $b + 3$ ,  $b^2 - 9$ ,  $-3b$ ,  $2b - 1$ , and 5.

$((4), (4) + 3, (4)^2 - 9, -3 \cdot (4), 2(4) - 1, \text{ and } 5) = 4, 7, 7, -12, 7, \text{ and } 5$ . The mode of a set of numbers is the number that appears in the set the most. The mode is 7 since it appears three times.

### 7 Answer A Difficulty Level 9

**11) When solving problems where one of the answers must be true, try to find one thing that makes each answer choice false.** If two numbers  $g$  and  $h$  average zero, do they both have to be zero? One example of the statement A)  $g = 0, h = 0$  would be false if  $g = 5$  and  $h = -5$  (or any additive inverse). The average of  $g$  and  $h$  would still be zero, which means  $g$  and  $h$  do not both have to be zero for the statement to be true. Using the exact two integers,  $g = 5$  and  $h = -5$  as an example would negate choice B) as that would make  $gh$  equal to  $5 \cdot -5 = -25$ . These exact two integers would also prove C) is not always true;  $5 - (-5) \neq 0$ . Any two numbers that average zero must also have a sum of zero. The answer must be D) because  $g + h$  must equal 0 for their average to equal zero. Answer E) can be easily eliminated if, for example,  $g = 0$  and  $h = 0$ .

### $g + h = 0$ Answer D Difficulty Level 7.5

**12)** Find the mean of a set of numbers by adding the numbers and dividing by the number of terms.  $4 + 8 + 12 + 0 + 1 = (25)$  and  $25 \div 5 = 5$

### 5 Answer A Difficulty Level 1

**13)** *When solving problems where one of the answers must be true, try to find one thing that makes each answer choice false.* Starting with answer choice A, find one example where every number in a set of five integers does not have to be a 5 to have a mean, median, and mode of 5. One example is {3, 5, 5, 5, 7}. The mean median and mode are still 5, but every number in the set is not a 5. To prove answer choice C is false, use a different set of numbers, {2, 5, 5, 6, 7} where the mean, median, and mode are still 5 but the median of the other numbers in the set; 2, 6, 7 is 6. Answer choice D can be proven false by the same example set used {2,5,5,6,7} before. This set has a mean, median, and mode of 5 but has only 2 5's in the set. Answer choice E can be proven False by the following set; {1,2,5,5,12} where the mean median and mode are still 5, but the largest integer 12 exceeds 11. Answer choice B cannot be proven False and is, therefore, the correct answer. None of the three examples used to make the other answer choices false, {3, 5, 5, 5, 7}, {2, 5, 5, 6, 7}, {1, 2, 5, 5, 12} can be used to make answer choice B false. In each case, the mean of the numbers in the set other than the number five is 5, which must be true to ensure the mean of the five integers will be 5.

The mean of the numbers in the set other than 5 is equal to 5

[Answer B](#) Difficulty Level 9

**14)** *When given an average, immediately multiply the average by the number of terms/subjects.* Working backward from the average formula makes these types of problems easy. Remember, Arithmetic mean is just a fancy term for the mean, which is a fancy term for average. The average of five integers is 15; Multiply;  $5 \times 15 = 75$ . The sum of all five integers is 75. Four of the integers are 18, 19, 20, and 21. Add the four integers, and the sum is 78, which is larger than the sum when a fifth number is added. Since five numbers sum (add) to 75, the last integer must be  $-3$ . ( $78 + (-3) = 75$ ).

From a visual perspective,

$\{18, 19, 20, 21, ?\}$

the four integers sum to 78. Since all five integers sum to 75, “?” must =  $-3$ .

$$(75 - 78 = -3)$$

$-3$  [Answer A](#) Difficulty Level 7.5

**15)** *When given an average multiply the average by the number of terms/subjects.* Working backward from the average formula makes these types of questions easy. The arithmetic mean (average) of seven integers is 8; Multiply;  $7 \times 8 = 56$ . The sum of the seven integers in the set is 56, where four of those integers sum (add) to  $(8 + 9 + 10 + 11 = 38)$ . The sum of the other three integers must be  $56 - 38 = 18$ . The average of three numbers that sum to 18 is  $18/3 = 6$ .

6 [Answer B](#) Difficulty Level 7.5

**16)** When given variables in this type of question, each number is separated by a comma. The first number is  $a + 9$ , the second is  $3a - 3$ , the third is 8, the next is  $2a + 8$ , and the fifth and last one is  $4a - 2$ . The five numbers have an average of  $3a - 1$ . To find the missing variable ( $a$ ), set up an average equation, add the numbers, and divide by the total. Since there are five numbers, add them up and divide by 5 to get their average. Set that equal to their given average.

$= \frac{a+9+3a-3+8+2a+8+4a-2}{5} = 3a - 1$ ; add like terms  $\frac{10a+20}{5} = 3a - 1$ ; Cross multiply and distribute the 5;  $15a - 5 = 10a + 20$ , subtract  $10a$  from both sides of the equation and add 5 to both sides.  $5a = 25$ , divide both sides by 5,  $a = 5$ .

(Another option is to simplify the left-hand side of the equation and not cross multiply)

$2a + 4 = 3a - 1$ , subtract  $2a$  from both sides and add 1 to both sides,  $a = 5$  (same result). Now substitute (plugin)  $a = 5$  into each expression;  $(5) + 9$ ,  $3(5) - 3$ , 8,  $2(5) + 8$ ,  $4(5) - 2$ . The five numbers are 14, 12, 8, 18, 18. **To find the median of a set of numbers put them in order from least to greatest. Cross out**

the same number of numbers from the far left and far right sides until only one or two numbers are left. If two numbers are left, add them and divide the result by two to get the median. If only one number is left, then that number is the median. In this set {8, 12, 14, 18, 18} ordered from least to greatest shows the number in the middle (**median**) is 14. The **mode** is the number in the set that occurs most often, which in this question is 18. The sum of the median and the mode =  $14 + 18 = 32$ .

32 [Answer E](#) Difficulty Level 9

17) To find the average add up the terms and divide by the number of terms:  

$$\frac{x^2+3x^2+6x^2+10x^2}{4} = \frac{20x^2}{4} = 5x^2$$
 (Remember: when adding polynomials add like terms). When dividing, only like terms divide so  $\frac{20}{4}$  reduces to 5.)

$5x^2$  [Answer A](#) Difficulty Level 7

18) The chart used for this question and the next is a frequency chart. The mode is the number or data points that occur most often. Since seven students correctly answered nine questions and 7 was the highest number of students for any number of questions answered correctly, 9 is the mode of the set.

9 [Answer E](#) Difficulty Level 3

19) To find the median, line up the data from least to greatest. Since multiple students have identical scores, do not forget to list all the students' scores.  
 {5, 6, 6, 6, 6, 7, 7, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 10, 10} Find the term in the middle by counting or crossing off the same amount of numbers on each end. If there are two numbers in the middle remaining like in this question, take their average. The median of this set is 9.

9 [Answer E](#) Difficulty Level 5

20) These types of questions are time killers on the ACT. Try each answer choice and try to see which changes the median of the set.

Answer choice A, adding three integers with the same median as the set will not change the median.

Answer choice B, adding two integers that are the same as the median will not change the median of the set.

Answer choice C, adding the same amount of terms on each side of the median will not change the median.

Answer choice D will only make the largest number larger, and the smallest smaller but will not change the median.

Answer choice E is correct because doubling every integer except the highest and lowest will change the median.

Example; {1, 2, 3, 4, 5} has a median of 3; becomes {1, 4, 6, 8, 5} ordered from least to greatest, {1, 4, 5, 6, 8} has a median of 5.

Doubling every integer except the highest and lowest

[Answer E](#) Difficulty Level 8

21) To solve this question correctly, first set up the given equation for the average or arithmetic mean. To find the arithmetic mean, add the terms and divide

by the number of terms:  $\frac{x+x+2+2x-4+3x-11+\frac{1}{2}x+3}{5} = 7$ , combine like terms  $\frac{\frac{15}{2}x-10}{5} =$

$7$ , multiply both sides of the equation by 5,

$\frac{15}{2}x - 10 = 35$ ,  $\frac{15}{2}x = 45$ ,  $x = 6$ . Finally, find the numbers by substituting  $x = 6$ :

6, 8, 8, 7, 6. To find the median of the numbers, line them up from least to greatest and find the one in the middle. 6, 6, 7, 8, 8; the median is 7.

7 [Answer B](#) Difficulty Level 9

22) If the average of  $a$ , 20, 40, and 80 is 45, set up an equation to find the given average:  $\frac{a+20+40+80}{4} = 45$ ,  $a + 140 = 180$ ,  $a = 40$ . To find  $b$ , set up an equation to find the given average:  $\frac{b+20+40+80}{4} = 60$ ,  $b + 140 = 240$ ,  $b = 100$ . The average of  $a$  and  $b$  is  $\frac{40+100}{2} = 70$

70 Answer A Difficulty Level 8

23) To find  $q$ , set up an equation to find the given average. If the average of 50, 70, and  $q$  is 80:  $\frac{50+70+q}{3} = 80$ ,  $120 + q = 240$ ,  $q = 120$ . To find  $r$ , set up an equation to find the given average. If the average of 30, 60,  $q$ , and  $r$  is 85:  $\frac{30+60+q+r}{4} = 85$ ,  $210 + r = 340$ ,  $r = 130$ . Therefore,  $q + r = 120 + 130 = 250$ .

250 Answer A Difficulty Level 8